A Nonlinear Two-Path Regression Test in Factor Asset Pricing Models

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1 Literature Review

Fama and MacBeth, 1973. Fama French 3 and 5 factors. GMM 1982. Ghysel's critique of the dynamic CAPM. Also should look at the local linear and local maximum likelihood estimation methods. **jagannathanAsymptoticTheoryEstimating1998** provides a good reading on the subject. **shankenEstimationBetaPricingModels1992** on the standard error of the Fama-Macbeth Regression.

For example, the CAPM assumes one factor that is the market excess return r_m.

$$E(r_i) = \beta_i E(r_m) \quad \text{where} \quad \beta_i := \frac{\text{cov}(r_i, r_m)}{\text{var}(r_m)}$$

Fama and MacBeth, 1973 derive testable restrictions from the CAPM model and embed the model in a linear regression:

$$r_{it} = \gamma_{0t} + \gamma_{1t}\beta_i + \gamma_{2t}\beta_i^2 + \gamma_{3t}s_i + \varepsilon_t$$

where β_i is estimated with $\hat{\beta}_i$ from time series regression for each i

$$r_{it} = \alpha + \beta_i r_{mt} + u_{it}$$

The testable hypotheses are

1. $E(\gamma_0) = 0$ there is no pricing error in the expected excess return.
2. $E(\gamma_2) = 0$ the linearity test.
3. $E(\gamma_3) = 0$ there is no other risk measure.
4. The Sharpe-Lintner Hypothesis that $E(\gamma_1) = R_{ft} > 0$.

Ma et al., 2020 considers **gibbonsTestEfficiencyGiven1989** tests with large N and points out that the factor loadings change substantially over time, they conduct a High-dimenisonal alpha test and

jegadeeshEmpiricalTestsAsset2019 and **chaiebTimeVaryingRiskPremia2018** are new papers that referenced the **jagannathanAsymptoticTheoryEstimating1998**, see also **jagannathanNoteAsymptoticCovariance1998**.

Generally, people test CAPM or other asset pricing models by testing that the $\alpha = 0$ (Jensen's α), meaning there is no systematic pricing error(in conditional expectation).

wangAssetPricingConditioning2003 does a testing of CAPM and factor models in a nonparametric setting with conditioning information. The starting point is the SDF characterisation of excess returns r_i:

$$E_t(m_{t+1}r_{i,t+1}) = 0 \implies E_t\left(\left[1 - r_{m,t+1}\frac{E_t(r_i, r_m)}{E(r_m^2)}\right]r_{i,t+1}\right) = 0 \tag{1}$$

Then $m_{t+1} = 1 - b(x)r_{m,t+1}$ is nonparametrically estimated from the market excess return $r_{m,t}, t = 1, \ldots, T$. The covariates x is four-dim in the empirical part. Then they test relationship by the following regression(they use weighted Least Square for technical reason).

$$\hat{m}_{t+1} r_{m,t+1} = z_t' \delta_i + u_{i,t+1} \quad \text{where} \quad z_t \in I_t$$

For testing a economic theory in a nonparametric setting, see Chapter 42.

For *error-in-variable* problem, there is on nonparametric regression with error in variables, they proposed deconvolution method to account for the observation errors. There is **bickelEfficientEstimationErrors1987** on efficient parametric regression: there is identification problem in the error-in-variable problem, which serves as an important example in Rothenberg. **reiersolIdentifiabilityLinearRelation1950** is a paper on the identification issue, it shows equivalence with the assumption of normal measurement error and show equivalence using characteristic functions, this method should be noted.

For *cross-sectional correlation problem*, we can see Li and Racine, 2007 chapter 18 and also **robinsonNonparametricTrendingRegression2012**.

acharyaAssetPricingLiquidity2005 proposes a liquidity factor and uses the cross-sectional method to verify. Further applications, see **jegadeeshEmpiricalTestsAsset2019** page 2. **jegadeeshEmpiricalTestsAsset2019** proposes an IV method to deal with the first step estimation error, or error in variable, this allows them to use individual stocks instead of portfolios, they do the test on CAPM **and** factor models.

Apparently Fama Macbeth didn't give statistical reasons for the two-pass regression procedure. **gibbonsTestEfficiencyGiven1989**, **gibbonsMultivariateTestsFinancial1982** and **shankenEstimationBetaPricingModels1992** should be necessary companion.

shankenEstimationBetaPricingModels1992 mentioned that the analysis of MLE is not easy because they have N, T both $\to \infty$. The underlying model is richer than just simply APT, but see Shanken 1982 1985 1992 on APT and **dybvigYesAPTTestable1985** for discussion.

shankenEstimationBetaPricingModels1992 gives a general parametric framework where he distinguished the factors that are and are not returns, and derive asymptotic distribution of the two-pass estimator of the γ's. The first lemma regards the first step

estimation.

See Ross for APT. **dybvigYesAPTTestable1985** says that APT theory incoporates assumptions other than a simple factor generating structure so any test of APT should take account of that. **chamberlainFundsFactorsDiversification1983** is also a must-read.

herskovicNetworksProductionAsset2018 referneced the **shankenEstimationBetaPricingModels199**

There is a different paper **shankenCurrentStateArbitrage1992**. For the **eubankTestingGoodnessFit19** might be useful for the inference. **hardleComparingNonparametricParametric1993**.

Follow the line of specification test as in **hardleComparingNonparametricParametric1993**, there is a paper **chenTestModelSpecification2008** which gives an extensive review of relevant literature. Two of the papers are worth mentioning, the genralized likelihood method and the test of $\hat{\varepsilon} = 0$. But it requires some ordering of the observations, which seems a bit odd. The empirical likelihood method and generalized likelihood ratio test seems to be important in nonparametric inference literature, should check it out. **fanSieveEmpiricalLikelihood2004** puts an even more general likelihood ratio where the error term is no longer required to be iin a parametric family.

There seems to be a lot of specificaton tests. and Fama-Macbeth is quite popular, **fengTamingFactorZoo2017** is an example, where they use double Lasso selection to both select the relevant factor and keep some of irrelevant factors in the regression to avoid the omitted variable problem and then they use Fama-macbeth to evaluate the validity of the factors.

hongConsistentSpecificationTesting1995 is a specification test with series regression.

fanNonlinearTimeSeries2008 also has a chapter on generalized likelihood ratio statistic.

Apparently **horowitzAdaptiveRateOptimalTest2001** is an important article, it's also mentioned in Li and Racine, 2007.

Here the econometric problem is to extend the GLR test to incorporate first-step estimation error.

I think some work need be done for the estimation of the covariance because there seem to be cross-sectional correlation. We can show asymptotic normality, see Li and Racine, 2007 chapter 2 and 18, should also check chapter 19 and 20. Then we follow **hardleComparingNonparametricParametric1993** to see whether the linearity holds,

which proposes a L^2 test and suggest bootstrap method. For that purpose, we need to find uniform convergence rate, see , which is based on **komlosApproximationPartialSums1975**, see **berkesKomlosMajorTusnady2014** for the extension to dependent data. **komlosApproximationPartial** states a result due to Bartfai 1966, and later a simple proof for that is proved in **erdNewLawLarge1970**.

As pointed out by Fama and MacBeth, 1973, we should calculate that $Eg(\cdot)$ is a linear function rather then $g(\cdot)$ being linear.

okuiKernelEstimationPanel2019 provides a kernel density estimation method with the first-step estimation error. **dhaeneSplitpanelJackknifeEstimation2015** provides a way to correct for biases.

Fama and Macbeth mentioned that the estimated $\hat{\beta}_i$ tends to be above the true β_i for large β, I wonder why that's the case.

cochraneAssetPricing2001 provides a way to put the two passes in one pass with GMM framework. **taoUniformTrinityTests2020** might also be helpful.

$$R_{it} = h_t(\beta_i) + e_{it}$$

where we estimate β_i from the time series of returns. Then each period we have h_t is only driven by the cross-section size and the time series estimation error in β_i. We want to test that $h_t(b) = b * R_m t$ for some observed $R_m t$ or even some latent market factor so $b * U_t$, since then we would have $h_t(b)/h_s(b) = U_t/U_s$ which does not vary with b, which would be testable

In section 3.5 of **gagliardiniTimeVaryingRiskPremium2016**, they propose a specification test and mentions another paper **kanPricingModelPerformance2013** which describes the asymptotic distribution of R^2 statistic and comparison between different asset pricing models' R^2.

fanNonparametricInferenceGeneralized2007 is a review article on GLR. It also mentions some potential questions in that area. I wonder if it's possible to combine **horowitzAdaptiveRateOptimalTest2001** with GLR.

escancianoAsymptoticDistributionfreeTests2018 seems to consider the case when there is stationary and mixing dependent data. I wonder if it applies to the case when we have cross-sectionally correlated data.

2 Introduction and Model Set-up

Let $(\Omega, \mathcal{H}, P)$ be a probability space, let $F(\omega) : \Omega \longrightarrow \mathbb{R}^k$ representing k candidate factors, and $\varepsilon(\iota, \omega) : [0,1] \times \Omega \longrightarrow \mathbb{R}$ be measurable. Assume there is a measure-preserving and ergodic time-shift transformation $S : \Omega \longrightarrow \Omega$ such that $S^t(\omega)$ is the time t state if the initial state $S^0(\omega) = \omega$. Define $F_t(\omega) = F(S^t(\omega)), \varepsilon_t(\iota) = \varepsilon(\iota, S^t(\omega))$.

Assume the time series dynamic of the excess returns $Y_t(\iota) = Y(\iota, S^t(\omega))$ follows

$$Y_t(\iota) = a(\iota) + b(\iota)' F_t + \varepsilon_t(\iota) \tag{2}$$

where $[a(\iota), b(\iota)']' : ([0,1], \mathcal{B}) \longrightarrow \mathbb{R} \times \mathbb{R}^k$ represents **time-invariant** factor loadings. (We can extend it to time-varying settings as in Gagliardini, Ossola, and Scaillet, 2016).

Assumption 1 (Random Sampling). *We observe a sample with sample size n by independently drawing $\left\{\iota_j : j \geq 1\right\}$ according to a probability distribution G on $[0,1]$. Assume ι_j are also defined on Ω and are independent of (a, b),F_t and ε_t.*

We want to test the null H_0 against the alternative H_1,

$$H_0 : E(Y(\iota) \mid b(\iota)) = b(\iota)'\gamma \quad a.s. \quad \text{for some} \quad \gamma \in \Gamma \subset \mathbb{R}^k \tag{3}$$

$$H_1 : P(E(Y(\iota) \mid b(\iota)) = b(\iota)'\gamma) < 1 \quad \text{for all} \quad \gamma \tag{4}$$

Assumption 2 (Model under H_0). *In addition to the model in Equation 2, we make the following assumptions on the data generating process*

1. $\beta = \left[\beta_1, \ldots, \beta_N\right]'$ *is independent of* F_t *for all* t.
2. $\varepsilon_t = (\varepsilon_{1t}, \ldots, \varepsilon_{Nt})'$ *are independent over time* t. $E\varepsilon_t\varepsilon_t' = \Sigma$, *there may be cross-sectional correlation. Assume that* $\|\Sigma\| < \infty$.
3. F_t *are stationary and* $E(\varepsilon_t \mid \beta, F_t) = 0$.

3 Log

3.1 2020-09-01

$$\hat{u}_i = \bar{Y}_i - \hat{b}_i'\hat{\gamma} = \bar{\varepsilon}_i - \left(\hat{b}_i - b_i\right)'\bar{\gamma} - \hat{b}_i'(\hat{\gamma} - \bar{\gamma}) \tag{5}$$

Roughly $\bar{\varepsilon}_i$ will be kept, the last term $\bar{\gamma} - \hat{\gamma}$ is of smaller order, the second term will contribute.

Let $R_2 = \left(\hat{b}_i - b_i\right)' \bar{\gamma}$,

Lemma 1 (R_2). $R_2 = \ldots$

Proof.

$$R_2 = \left(\Sigma_F^{-1} + o_p(1)\right) \frac{1}{T} \sum_t \varepsilon_{it} \left(F_t - \bar{F}\right) \tag{6}$$

$$= \frac{1}{T} \sum_t \varepsilon_{it} \Sigma_F^{-1} \left(F_t - \bar{F}\right) \bar{\gamma} \tag{7}$$

$\Sigma_F^{-1}\left(F_t - \bar{F}\right)\bar{\gamma} = \Sigma_F^{-1}\left(F_t - \bar{F}\right)\left(\gamma + \bar{F} - EF\right)$, let $c_t = \Sigma_F^{-1}\left(F_t - \bar{F}\right)\gamma$. We will have

$$\hat{u}_i = \frac{1}{T} \sum_t (1 - c_t)\varepsilon_{it} + R_3 + R_4 \tag{8}$$

where $R_3 = \hat{b}_i'(\bar{\gamma} - \hat{\gamma})$, R_4 comes from the estimate of c_t.

c_t is independent of ε_{it} for all i, t. □

We construct based on the following statistic:

$$\mathbf{T}_{n,T} = \frac{1}{n(n-1)} \sum_i \sum_{j \neq i} \hat{u}_i \hat{u}_j \frac{1}{h} K\left(\frac{\hat{b}_i - \hat{b}_j}{h}\right) \tag{9}$$

we show that under scaling it converges in distribution.

The draft is

$$\mathbf{T}_{n,T} = \frac{1}{n(n-1)T^2} \sum_t (1 - c_t) \sum_{i \neq j} \left[\varepsilon_{it}\varepsilon_{jt} \frac{1}{h} K\left(\hat{b}_i - \hat{b}_j / h\right)\right] \tag{10}$$

$$= \frac{1}{n(n-1)T^2} \sum_t (1 - c_t) \sum_{i \neq j} \left[\varepsilon_{it}\varepsilon_{jt} \frac{1}{h} \left(K\left(b_i - b_j / h\right)\right) + R_K\right] \tag{11}$$

$$= \frac{1}{\sqrt{n(n-1)}T^2} \sum_t (1 - c_t) \frac{1}{\sqrt{n(n-1)}h} \sum_{i \neq j} \varepsilon_{it}\varepsilon_{jt} K\left(\frac{b_i - b_j}{h}\right) \tag{12}$$

My guess is that

$$nT^{\frac{3}{2}}\mathbf{T} \rightsquigarrow N(\mu, \mathrm{var}) \tag{13}$$

3.2 2020-09-11

We break down $\hat{u}_i$:

$$\hat{u}_i = \bar{\varepsilon}_i - \left(\hat{b}_i - b_i\right)' \bar{\gamma} - \hat{b}_i'(\hat{\gamma} - \bar{\gamma}) \tag{14}$$

$$= \bar{\varepsilon}_i - R_{b,i}'\bar{\gamma} - \left(b_i + R_{b,i}\right)' \left(\hat{B}'\hat{B}\right)^{-1} \hat{B}' \left[\bar{\varepsilon} - (\hat{B} - B)\bar{\gamma}\right] \tag{15}$$

We can calculate the second remainder term:

$$\sqrt{NT}\hat{b}_i'(\hat{\gamma} - \gamma) = \hat{b}_i'\left(\Sigma_B^{-1} + o_p(1)\right) \frac{1}{\sqrt{T}} \sum_t \frac{1}{\sqrt{N}} \sum_j \left(b_j(1 - c_t)\varepsilon_{jt}\right) = O_p(1) \tag{16}$$

Hence $\hat{u}_i = \frac{1}{T}\sum_t \varepsilon_{it}(1 - c_t) + O_p(\frac{1}{\sqrt{NT}})$, where the last term is uniform over all i. Then we can calculate $\sum_{i \neq j} \hat{u}_i \hat{u}_j K_h\left(\hat{b}_i - \hat{b}_j\right)$.

$$\mathbf{T}_{n,T} = \frac{1}{n(n-1)} \sum_i \sum_{j \neq i} \hat{u}_i \hat{u}_j \frac{1}{h} K\left(\frac{\hat{b}_i - \hat{b}_j}{h}\right) \tag{17}$$

We want to calculate var $\left(\mathbf{T}_{n,T}\right)$ and determine the scaling.

3.3 2020-10-16

The model is

$$Y_{it} = a_i + b_i' F_t + \varepsilon_{it}$$

and we have under the H_0,

$$E(Y_{it}) = b_i'\gamma$$

So $a_i = b_i'\gamma - b_i' E(F_t)$, that is

$$Y_{it} = b_i'(F_t + \gamma - E(F_t)) + \varepsilon_{it} \quad \text{for all} \quad i$$

3.3.1 Fist step estimation

We have $\hat{b}_i = \left(F_D' F_D\right)^{-1} F_D' Y_{i,D}$, because $\bar{Y}_i = a_i + b_i'\bar{F} + \bar{\varepsilon}_i$ we have $Y_{i,D} = Y_i - \mathbf{1}\bar{Y}_i = F_D b_i + \bar{\varepsilon}_i$ and hence

$$\hat{b}_i = b_i + \left(F_D' F_D\right)^{-1} F_D' \bar{\varepsilon}_i$$

3.3.2 Second Step estimation

We stack $\hat{B} = \left[\hat{b}_1, \ldots, \hat{b}_n\right]'$. In the cross-sectional,

$$\bar{Y}_i = a_i + b_i'\bar{F} + \bar{\varepsilon}_i = b_i'\gamma + b_i'\left(\bar{F} - EF\right) + \bar{\varepsilon}_i$$

Let $\bar{\gamma} = \gamma + \bar{F} - EF$, then we can estimate it by

$$\hat{\bar{\gamma}} = \left(\hat{B}'\hat{B}\right)^{-1}\hat{B}'\bar{Y} = \bar{\gamma} + \left(\hat{B}'\hat{B}\right)^{-1}\hat{B}'\bar{\varepsilon} = \bar{\gamma} + R_\gamma$$

And the residual we use for testing is

$$\begin{aligned}
\hat{u}_i &= \bar{Y}_i - \hat{b}_i'\hat{\bar{\gamma}} && (18)\\
&= b_i'\bar{\gamma} + \bar{\varepsilon}_i - \hat{b}_i'\hat{\bar{\gamma}} && (19)\\
&= \bar{\varepsilon}_i - \left(\hat{b}_i - b_i\right)'\bar{\gamma} - \hat{b}_i'\left(\hat{\bar{\gamma}} - \bar{\gamma}\right) && (20)\\
&= \bar{\varepsilon}_i - R_{b,i}'\bar{\gamma} - \left(b_i + R_{b,i}\right)'R_\gamma && (21)
\end{aligned}$$

The test statistic is based on scaling the following statistic:

$$\mathbf{T}_{n,T} = \frac{1}{n(n-1)}\sum_i\sum_{j\neq i}\hat{u}_i\hat{u}_j\frac{1}{h}K\left(\frac{\hat{b}_i - \hat{b}_j}{h}\right) \tag{22}$$

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